

MTL1002: TUTORIAL 2
(LINEAR ALGEBRA)

- (1) Let $v_1 = (1, 2)$ and $v_2 = (0, 1) \in \mathbb{R}^2$.

(a) Describe the following subsets of $\mathbb{R}^2$ geometrically.

$$\begin{aligned} W_1 &:= \{tv_1 : t \in \mathbb{R}\} \\ W_2 &:= \{tv_2 : t \in \mathbb{R}\} \\ W_3 &:= \{sv_1 + tv_2 : s, t \in \mathbb{R}\} \\ W_4 &:= \{sv_1 + tv_2 : 0 \leq s, t \leq 1\}. \end{aligned}$$

(b) Which of the above subsets W_1, W_2, W_3, W_4 are subspaces of $\mathbb{R}^2$? Justify your answer in each case.

(c) Show that $\{v_1, v_2\}$ is a linearly independent subset of $\mathbb{R}^2$.

(d) Let $v_3 = (2, 3) \in \mathbb{R}^2$. Is $\{v_1, v_2, v_3\}$ linearly independent?

- (2) Consider $\mathbb{C}^2$ as the vector space over $\mathbb{C}$ under component-wise addition and usual scalar multiplication.

(a) Show that $\{(1 + i, 2), (2, 1)\}$ is linearly independent.

(b) Show that $\{(1, 2), (0, i), (i, 1 - i)\}$ is linearly dependent.

(c) Show that every element in $\mathbb{C}^2$ (which are ordered pairs) can be written as a linear combination of $v_1 = (1 + i, 2)$ and $v_2 = (2, 1)$. Moreover, prove that up to change of order (of v_1 and v_2) such a linear combination is unique (for each ordered pair).

(d) Show that every element in $\mathbb{C}^2$ can be written as a linear combination of $v_1 = (1, 2)$, $v_2 = (0, i)$, $v_3 = (i, 1 - i)$ in more than one way.

- (3) Consider $\mathbb{C}^2$ as a vector space over $\mathbb{R}$.

(a) Show that the following subset X is linearly independent:

$$X = \{(1 + i, 1 - i), (1 - i, 1 + i), (2, i), (3, 2i)\}.$$

(b) For $a, b, c, d \in \mathbb{R}$, express $(a + ib, c + id) \in \mathbb{C}^2$ as an $\mathbb{R}$ -linear combination of vectors belonging to X .

- (4) Let V be a vector space over $\mathbb{F}$ and $u, v, w \in V$. Show that

$$\{u, v, w\} \text{ is linearly independent} \Leftrightarrow \{u + v, v + w, w + u\} \text{ is linearly independent.}$$

- (5) Let V be the vector space of all real valued continuous functions on the interval $[0, 1]$ over $\mathbb{R}$. Let $f_1, f_2, f_3 \in V$ be defined by $f_1(x) = \sin x$, $f_2(x) = \cos x$, $f_3(x) = e^x$. Is the subset $\{f_1, f_2, f_3\}$ linearly dependent?

- (6) (a) Consider $\mathbb{R}^3$ as a vector space over $\mathbb{R}$ with standard operations. Find the coordinates of the vector $(a, b, c) \in \mathbb{R}^3$ relative to the ordered basis

$$\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}.$$

- (b) Consider the vector space $\mathcal{P}_3$ of polynomials of degree at most 2 with coefficients from $\mathbb{R}$. Find the coordinates of $a + bX + cX^2$ with respect to the ordered basis

$$\{1, 1 + X, 1 + X^2\}.$$

- (c) Consider $\mathbb{R}^3$ as a vector space over $\mathbb{R}$ with standard operations. Find the coordinates $(a, b, c) \in \mathbb{R}^3$ with respect to the following bases

$$B_1 = \{(1, 2, 1), (1, 2, 3), (0, 1, 1)\},$$

$$B_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}.$$

Also, write the change of coordinate matrix.

- (7) Let V be a vector space over $\mathbb{F}$.

- Show that if $v \in V$ then $\mathbb{F}v := \{\lambda v : \lambda \in \mathbb{F}\}$ is a subspace of V .
- Show that if W_1, W_2 are subspaces of V , then $W_1 \cap W_2$ is a subspace of V .
- Show that the intersection of any collection of subspaces of a vector space is also a subspace.
- Suppose W_1 and W_2 are subspaces of a vector space V . Show that $W_1 \cup W_2$ is a subspace of V if and only if either $W_1 \subset W_2$ or $W_1 \supset W_2$.
- Let X be a nonempty subset of a vector space V over $\mathbb{F}$. Let

$$\text{span}(X) := \left\{ \sum_{i=1}^n a_i v_i : n \in \mathbb{N}, a_i \in \mathbb{F}, v_i \in X \right\}$$

and let $\langle X \rangle$ be the intersection of all the subspaces of V which contain X . Show that $\text{span}(X)$ and $\langle X \rangle$ are subspaces of V . Also show that $\text{span}(X) = \langle X \rangle$.

- (8) In each of the following cases show that $W_1 + W_2 = V$ (directly) and find $\dim(W_1 \cap W_2)$. Verify the formula

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

- $V = \mathbb{R}^2$, W_1 is the X -axis, W_2 is the Y -axis.
- $V = \mathbb{R}^2$, W_1 and W_2 are distinct lines passing through the origin.
- $V = \mathbb{R}^3$, W_1 is the XY -plane and W_2 is the YZ -plane.
- $V = M_n(\mathbb{R})$, $W_1 = \{A \in M_n(\mathbb{R}) : A \text{ is upper triangular}\}$, $W_2 := \{A \in M_n(\mathbb{R}) : A \text{ is lower triangular}\}$.
- $V = M_n(\mathbb{R})$, where W_1 is the space of $n \times n$ symmetric matrices and W_2 is the space of $n \times n$ skew-symmetric matrices.

- (9) Which of the following functions is a linear transformation? Justify.

- $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T_1(x, y) = (x^2 + y^2, x - y)$ over $\mathbb{R}$
- $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T_2(x, y) = (x + y + 1, x - y)$ over $\mathbb{R}$
- $T_3 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T_3(x, y) = (ax + by, cx + dy)$ over $\mathbb{R}$
- $T_4 : \mathbb{C} \rightarrow \mathbb{C}$ define by $T(z) = \bar{z}$ over $\mathbb{C}$
- Rotation about the origin by an angle θ in $\mathbb{R}^2$ (write an expression for rotation).
- $T_5 : M_{m \times n}(\mathbb{F}) \rightarrow M_{n \times m}(\mathbb{F})$ defined by $T_5(A) = A^t$ (A^t is the transpose of A)
- $T_6 : M_n(\mathbb{F}) \rightarrow \mathbb{F}$ defined by $T_6(A) = \text{tr}(A)$
- $T_7 : \mathcal{P}_n \rightarrow \mathcal{P}_n$ such that $T_7(p)(x) = p(x - 1)$
- $T_8 : \mathcal{P}_n \rightarrow \mathcal{P}_{n+1}$ such that $T_8(p)(x) = xp(x) + p(1)$